

ESTIMATING THE EFFORT TO DEVELOP SCREEN MOCKUP

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Definition

- **GOAL - Analyze:** the development of screen mockups starting from use cases ***For the purpose of:*** building a prediction model ***With respect to:*** the effort ***From the point of view of:*** researchers and requirements analysts ***In the context of:*** small/medium sized software projects and young developers and professional software engineers
- **Research Question (RQ)** - Are the estimates based on the prediction models built using use case size measures better than those provided by the baseline benchmark when predicting the effort to develop screen mockups?



Outline

- Motivation
- Empirical study
 - Design
 - Results and Discussion
- Future work

Motivation



- Usefulness of screen mockup to reduce ambiguities of FUR and to improve communication among stakeholders in requirement engineering process

Use Case: Registering a Supply

Level: User-Goal

Intention in Context: The Secretary wants to record the supply of products. The products are always paid in cash upon their arrival.

Primary Actor: Secretary

Main Success Scenario:

1. The Secretary provides as the input the number of boxes and the type of products to be bought.
2. The system shows how much money are available in the cash (i.e., the daily availability of the cash).
3. If there is enough money, the system confirms the supply and shows the current situation of the available products.
4. The use case ends successfully.

Extension:

- 3.a. If there is not enough money, the system informs the Secretary that the required operation cannot be accomplished. The use case fails.

Rifornimento Cialde

Ordine Scatole

Numero scatole: Tipo:

Carrello

Tipo	Quantità Scatole	Costo
Caffè	2	155 €
Caffè Arabica	1	77,5 €
The	1	77,5 €

Saldo Cassa: 200 €

Situazione Attuale Prodotti

Tipo	Quantità Bustine
Caffè	390
Caffè Arabica	150
The	60
Camomilla	37

Motivation (2)



- Few studies on the empirical assessment of screen mockups
 - Ref. [3]: an improvement in the comprehension of FURs ranging from 26% to 46% when use cases are enriched with screen mockups
 - Ref. [4]: participants perceived screen mockup construction as a non-onerous activity and it requires low effort as compared to use case writing
- Screen mockups are a viable means in the requirements engineering field

But...

How much effort is needed to develop mockups and what an estimation model needs to predict the cost to develop screen mockups?

Context selection

- **23 participants:** 12 students in Computer Science at the University of Basilicata and 11 young practitioners
 - Students passed a basic SE course before participating to our study (next generation of young professional developers)
 - The professional developers have been working for software companies of the contact network of one of the authors
- Not graded on the experimental tasks
- **4 desktop applications (University of Genova):** availability of requirement specification documents, included use cases compliant with SWEED template



Use Case: Registering a Supply Level: User-Goal Intention in Context: the supply of products: cash upon their arrival. Primary Actor: Secretary Main Success Scenario: 1. The Secretary provides the supply of products. 2. The system shows how much money are available in the cash (i.e., the daily availability of the cash). 3. If there is enough money, the system confirms the supply and shows the current situation of the available products. Extension:	Use Case: Registering a Supply Level: User-Goal Intention in Context: The Secretary wants to record the supply of products. The products are always paid in cash upon their arrival. Primary Actor: Secretary Main Success Scenario: 1. The Secretary provides as the input the number of boxes and the type of products to be bought. 2. The system shows how much money are available in the cash (i.e., the daily availability of the cash). 3. If there is enough money, the system confirms the supply and shows the current situation of the available products. 4. The use case ends successfully. Extension:	Use Case: Registering a Supply Level: User-Goal Intention in Context: The Secretary wants to record the supply of products. The products are always paid in cash upon their arrival. Primary Actor: Secretary Main Success Scenario: 1. The Secretary provides as the input the number of boxes and the type of products to be bought. 2. The system shows how much money are available in the cash (i.e., the daily availability of the cash). 3. If there is enough money, the system confirms the supply and shows the current situation of the available products. 4. The use case ends successfully. Extension:
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Selected variables

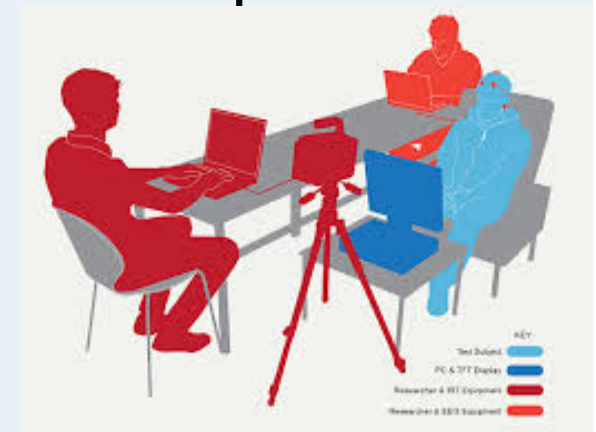
Measure	Description
#chars	the number of characters in the use case
#words	the number of words in the use case
#steps	the number of steps in the flow of events
#exts	the number of extensions in the flow of events
EFH	the effort required to define and design the screen mockups by analyzing the use case



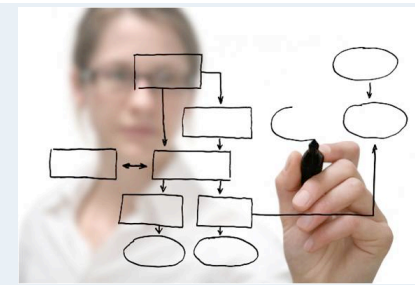
- We postulated that the effort to develop the connected screen mockup(s) should be proportional, on a first approximation, to the number of: **characters** and **words** in the use case, **steps** in the main success scenario, and **extension points** in the flow of events

Execution

- **Training session:** two screen mockups (Pencil tool) to develop
- **Filling in a pre-questionnaire:** to collect demographic info about the participants and previous work experience
- **Developing screen mockup task :** given a use case (from 3 to 8 steps and number of exts from 1 to 5), participants were asked to develop one or more mockups
- **Filling in a post-questionnaire:** to get qualitative insight about the task and the tool used to draw screen mockups
- Single experimental session
- Under the control of the same supervisor
- Participants not informed on RQ

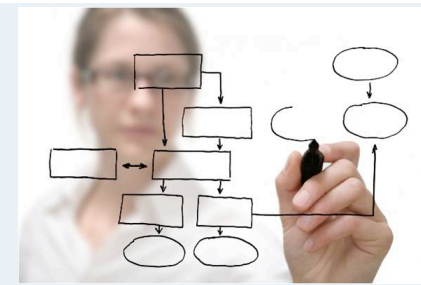


Data Analysis

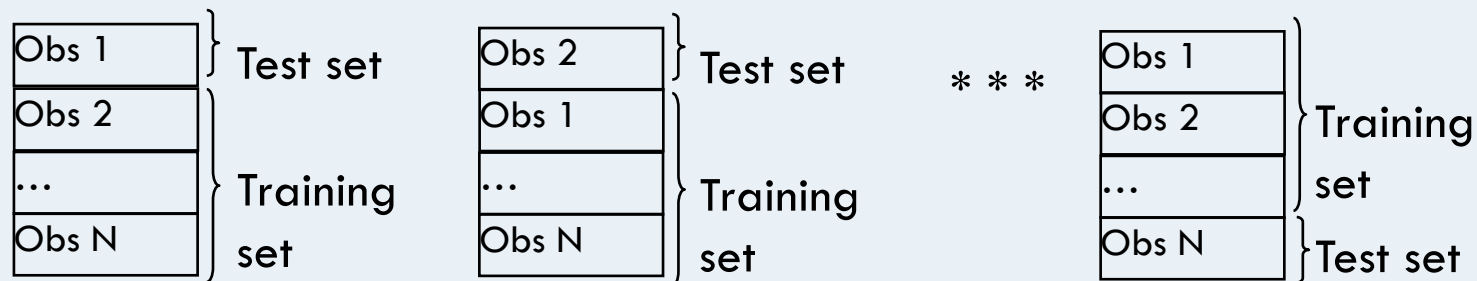


- **Estimation technique:** StepWise Linear Regression (SWLR)
 - $EFH = a_1 * chars + a_2 * steps + a_3 * exts + a_4 * words + b$
 - Independent variables selected in stages (automatic procedure in the statistical package R)
 - R^2 to evaluate the goodness of fit, showing the amount of variance of the dependent variable explained by the model related to the independent variables
- **Baseline benchmark:** The median of efforts (MedEFH) to develop screen mockups in the past
 - To verify the use of a easier approach before to consider more complex methods

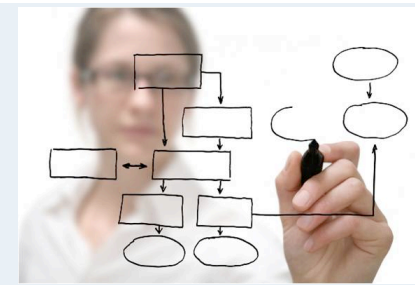
Data Analysis (2)



- **Validation method:** Leave-1-out cross validation
 - N different subsets of training (TS) and validation (VS) sets, where each validation set had one observation. TS used to build the estimation model and VS used to assess the obtained estimate



Data Analysis (3)



- ***Evaluation criteria:***

- Descriptive statistics (min, max, mean, median, StdDev) of absolute residuals $AR = | \text{actual effort} - \text{predicted effort} |$
- Correlation analysis (Kendall'tau and Spearman's rho) between predicted and actual efforts
 - Value in $[-1, 1]$: where +1 a perfect correlation, -1 a perfect inverse correlation. 0 no correlation
 - P-value: significance of the correlation between predicted effort and the actual effort
- Wilcoxon test: to verify whether the ARs obtained with SWLR models are significantly less (i.e., better) than ARs achieved with MedEFH
- For each test we decided to accept a probability of 5% of committing a Type-I-Error

Results



Data set	Variable Employed	R^2	Spearman's rho		Kendall's tau	
			statistic	p-value	statistic	p-value
Professionals	#chars #exts	0.853	0.900	< 0.01	0.745	< 0.01
Students	#chars #exts	0.871	0.895	< 0.01	0.727	< 0.01

- High R^2 values (> 0.8)
- #chars and #exts: best (significant) effort predictors
- Spearman's rho and Kendall's tau tests revealed that the predicted efforts are significantly correlated with the actual efforts
- Both the models provided good estimates for the effort to develop screen mockups from use cases

Results (2)



Data set	Estimation Technique	Min	Max	Mean	Med	St.Dev.
Professionals	SWLR	0.25	26.33	7.88	4.36	8.30
	MedEFH	1.00	33.50	12.36	10.50	11.05
Students	SWLR	0.63	10.44	5.94	5.44	3.08
	MedEFH	1.00	26.00	11.20	11.00	8.97

- Descriptive statistics of AR obtained with the leave-one-out cross validation show that our models outperform MedEFH (the baseline benchmark).
 - Results for Students and Professionals are close
- Wilcoxon test revealed that the AR obtained with SWLR models are significantly smaller (better) than those achieved with MedEFH (p-value=0.034 for professionals, p-value=0.046 for students)

Post Experiment Questionnaire

Q1-Q3: Questions about the task

Q4-Q7: Questions about the Pencil tool

Q8-Q12: Questions about the use of screen mockups added to use cases

ID	Questions
Q1	I had enough time to complete the task:
Q2	The used experimental documentation was clear:
Q3	I found useful the execution of the task:
Q4	The ease of use of the Pencil tool is:
Q5	Finding the widgets (e.g., combo box) in the Pencil's palette is
Q6	The use of standard drawing operations (e.g., aligning, ordering, scaling and rotating) in the Pencil tool is:
Q7	Specifying the features of the widgets (e.g., color) is:
Q8	The complexity of the screen mockups you developed is:
Q9	The effort to develop screen mockups with respect to the effort to develop use cases is
Q10	Adding existing use cases with screen mockups is:
Q11	I believe that adding use cases with screen mockups improves the comprehension of FURs:
Q12	How do you judge your experience in the development of screen mockups?

Possible Answers

Q1, Q2, Q3, Q11 → 1=strongly agree, 2=agree, 3=not certain, 4=agree, 5=strongly agree

Q4, Q8, Q12 → 1=very low, 2=low, 3=medium, 4=high, 5=very high

Q5, Q6, Q7 → 1=very difficult, 2=difficult, 3=medium, 4=easy, 5=very easy

Q9 → 1=very low, 2=low, 3=uncertain, 4=high, 5=very high

Q10 → 1=very burdensome, 2=burdensome, 3=uncertain, 4=not much burdensome, 5=not burdensome

Post Experiment Questionnaire (2)

- Participants considered appropriate the time to conduct the experiment (Q1). They clearly understood the documents (Q2), and found useful the tasks (Q3)
- Participants judged high or very high the ease of use of the Pencil tool for: finding the widgets in the palette (Q4), using the standard drawing operations (Q5), and specifying the widget features(Q6)
- The greater part of all the participants declared that the complexity of the developed screen mockups was medium (Q8). The participants did not have a clear-cut opinion on the difference between the effort to develop screen mockups and the effort to define use cases (Q9) and about the task of adding previously defined use cases with screen mockups

Post Experiment Questionnaire (3)

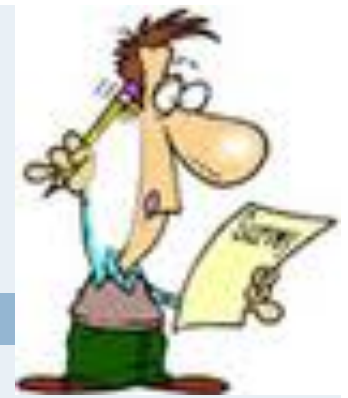
- The greater part of the participants did not agree on the fact that adding use cases with screen mockups improves the comprehension of FURs (Q1 1). This is in contrast with the results obtained in previous study [4]. The students judged medium-high their experience in designing/developing screen mockups (Q1 2) while professionals judged it medium-low

Main findings



- RQ can be positively answered: *the built models outperform the baseline benchmark*
- #charts and #exts resulted to be good predictors of the effort to develop screen mockups
- This preliminary result is particularly interesting from a practical point of view:
 - In a previous study [3], the comprehension of FURs increased when use cases added with screen mockups.
 - With the built SWLR models, we found a way to predict with good approximation the time to add use cases with screen mockups

Threats to validity



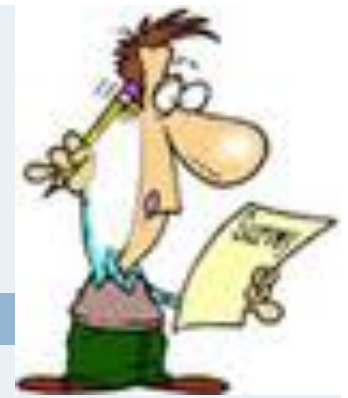
Internal validity

- Experimental procedure, Treatments, or Experiences of participants. Our best to control “extraneous factors”

External validity

- Use cases from 4 applications developed in University courses. Replications in other domains
- Use of a specific tool to develop screen mockup (Pencil)
- Students and professional results are close

Threats to validity (2)



Construct validity

- The measure for dependent variable well known and used
- Other independent variables could be used. However, our SWLR models are characterized by high R^2 values (> 0.8)

Conclusion validity

- Evaluation criteria allowed assessing in an objective way the effectiveness of the predictions
- Use of proper non parametric statistical test (Wilcoxon)
- Low number of participants. Replications with larger data sets